

HYDERABAD BEGUMPET, India

WMO: 431280

Lat: 17.452N

Lon: 78.461E

Elev: 531

StdP: 95.11

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.4	14.9	2.0	4.7	29.9	4.0	5.4	29.6	5.2	26.6	4.6	26.5	0.1	0	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.6	41.0	22.0	39.9	22.0	38.7	21.9	26.4	31.8	25.7	31.2	25.2	30.6	2.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	21.6	28.8	24.3	20.5	27.9	24.0	20.1	27.5	86.1	32.2	83.2	31.4	80.8	30.6	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.3	5.6	DB	10.4	42.7	1.6	1.5	9.3	43.8	8.3	44.7	7.4	45.5	6.3	46.6	
			WB	8.2	27.7	1.6	1.1	7.1	28.6	6.1	29.2	5.2	29.9	4.1	30.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	22.6	25.4	28.9	31.7	33.2	29.6	27.0	26.3	26.6	25.9	23.9	22.3	(6)
	DBStd	3.75	1.92	1.75	2.01	2.06	2.19	2.52	1.75	1.40	1.33	1.52	1.67	1.84	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	1	1	0	0	0	0	0	0	0	0	0	0	0	(9)
	CDD10.0	6189	392	431	584	652	720	589	528	506	498	492	416	382	(10)
	CDD18.3	3148	134	198	326	402	462	339	269	247	248	234	166	124	(11)
	CDH23.3	36206	1184	2144	4325	5937	7111	4272	2572	2027	2121	2061	1379	1073	(12)
	CDH26.7	17949	361	989	2458	3705	4694	2212	897	551	668	737	399	277	(13)
(14) Wind	WSAvg	2.4	1.6	1.9	1.9	2.0	2.7	3.8	3.9	3.4	2.4	1.8	1.6	1.4	(14)
(15) Precipitation	PrecAvg	823	5	8	17	23	36	108	167	183	165	97	17	6	(15)
	PrecMax	1384	51	81	166	99	233	225	422	469	484	324	58	94	(16)
	PrecMin	415	0	0	0	0	0	25	32	20	14	0	0	0	(17)
	PrecStd	197	11	18	29	23	43	52	89	101	99	77	18	16	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	36.2	40.1	42.2	43.0	41.0	34.8	33.0	33.8	33.8	32.0	33.1	(19)
		MCWB	19.5	18.7	20.7	21.2	22.2	22.7	22.6	23.1	24.1	21.6	19.7	19.5	(20)
	2%	DB	31.2	34.8	38.2	40.8	41.3	38.1	33.1	31.4	32.1	32.2	30.9	30.4	(21)
		MCWB	18.6	18.7	20.6	21.6	22.2	23.3	23.4	23.5	23.8	21.9	20.0	19.1	(22)
	5%	DB	30.0	33.2	37.0	39.2	40.2	36.2	32.0	30.5	31.1	31.2	29.9	29.1	(23)
		MCWB	18.2	18.9	20.1	21.8	22.1	23.3	23.5	23.5	23.8	21.6	20.0	18.3	(24)
	10%	DB	28.9	32.0	35.6	38.0	39.1	34.8	30.9	29.5	30.1	30.2	29.0	28.1	(25)
		MCWB	18.1	18.8	19.8	21.7	22.1	23.2	23.3	23.5	23.6	21.7	19.8	18.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.4	23.2	24.8	26.6	27.1	26.8	26.2	26.1	26.9	26.9	24.6	23.6	(27)
		MCDB	28.2	29.1	31.6	35.5	35.9	34.0	30.5	29.6	30.0	30.0	28.3	28.8	(28)
	2%	WB	21.2	22.1	23.9	25.2	25.8	26.0	25.3	25.3	25.9	25.5	23.2	22.2	(29)
		MCDB	26.4	27.9	30.9	33.7	34.4	32.5	29.4	28.6	29.3	28.7	27.6	26.8	(30)
	5%	WB	20.5	21.4	23.0	24.5	25.1	25.3	24.7	24.7	25.0	24.5	22.6	21.2	(31)
		MCDB	25.7	27.5	30.1	32.8	33.9	31.4	28.8	28.0	28.7	27.9	26.6	25.6	(32)
	10%	WB	19.8	20.6	22.3	23.8	24.4	24.7	24.3	24.3	24.4	23.7	22.0	20.4	(33)
		MCDB	25.3	27.2	29.9	32.1	33.5	30.6	28.4	27.6	28.0	27.1	25.6	24.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.3	13.6	12.5	11.6	8.9	6.8	6.4	7.2	9.0	10.9	12.7	(35)
		MCDBR	14.4	15.5	15.3	14.0	12.7	11.0	8.5	7.8	8.7	10.6	12.1	14.3	(36)
	5% WB	MCWBR	4.9	4.8	4.4	4.6	4.5	3.6	3.3	3.4	3.4	4.2	4.7	5.6	(37)
		MCWBR	11.5	11.7	12.6	12.4	11.4	8.9	7.1	6.6	7.4	8.1	8.5	10.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.451	0.463	0.531	0.571	0.605	0.583	0.540	0.522	0.515	0.531	0.478	0.455	(40)
		taud	2.077	2.062	1.907	1.858	1.785	1.877	2.006	2.067	2.081	1.993	2.077	2.080	(41)
	Ebn at Noon		826	839	794	763	728	737	771	791	797	768	794	807	(42)
		Edh at Noon	156	166	200	211	224	202	178	169	165	175	155	151	(43)
(44) All-Sky Solar Radiation	RadAvg		4.86	5.70	6.20	6.62	6.56	5.22	4.41	4.42	4.81	5.01	4.83	4.63	(44)
	RadStd		0.23	0.22	0.32	0.24	0.23	0.48	0.38	0.27	0.33	0.26	0.35	0.25	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.25	N/A	-1.67	+0.96	N/A	N/A	N/A	N/A	+140	+139
Regional (3 neighbors)	+0.23	N/A	N/A	+0.73	N/A	N/A	N/A	N/A	+140	+139

Nomenclature: See separate page